Supplementary Material

MAML-Select: An Online Adaptive Client Selection Method for Federated Learning via Meta-Learning

S1. Scope of the Supplement

This supplement reports additional evidence used during the revision. The main letter keeps the compact convergence and trade-off evidence, the λ-sensitivity plot, the feature-ablation plot, and the core tables. Here, we include the supporting material requested by the reviewers: all-dataset convergence views, paired tests, resource-reduction summaries, extended λ sensitivity, state-feature ablations, fairness and hardware-tier coverage, and selector-overhead scaling. All results use the same non-IID setting as the main manuscript: $N = 100$, $K = 10$, $E = 5$, batch size 32, Dirichlet $\alpha = 0.5$, and seeds 42, 123, and 2026 unless the table explicitly reports a smaller n. CIFAR-100 is analyzed at 150 rounds so that CriticalFL results are included without imputation.

Energy and carbon values are model-based estimates derived from client-tier cost and declared grid intensity.

S2. Additional Visual Evidence

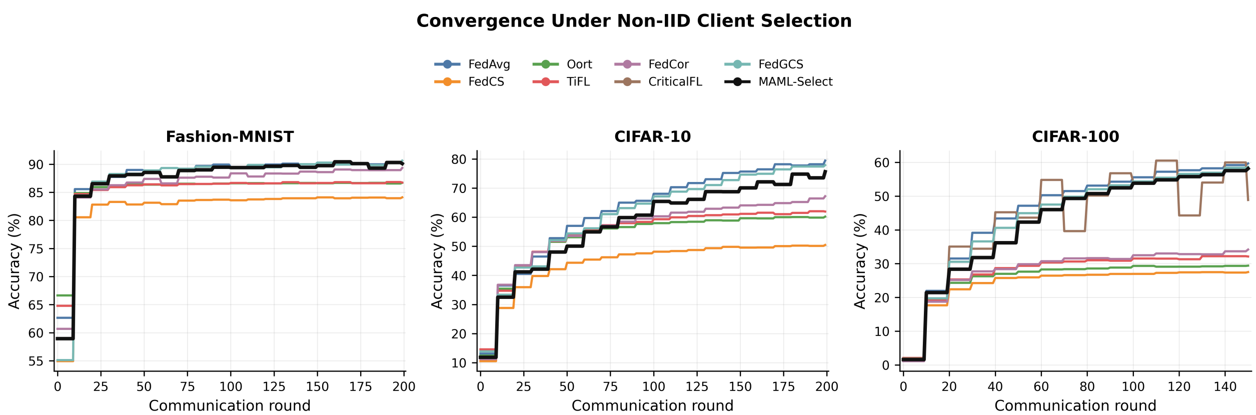


Figure 1: All-dataset convergence summary used to check consistency before curating the main-letter figures. The main paper reports the final CIFAR-100 four-panel version because it is the hardest setting and includes CriticalFL.

S3. MAML-Select Relative to FedAvg

Table 1: MAML-Select relative to FedAvg over shared seeds. Positive reductions indicate lower resource use for MAML-Select; ΔAcc. is signed.

Dataset	ΔAcc. (pp)	TFLOPs $\downarrow$ (%)	Energy $\downarrow$ (%)	Carbon $\downarrow$ (%)	n
Fashion-MNIST	-0.11	12.8	16.4	16.4	3
CIFAR-10	-3.81	21.5	24.7	24.7	3
CIFAR-100	-1.51	1.7	4.7	4.7	3

S4. Sensitivity, Ablation, and Overhead Tables

S5. Interpretation for the Revised Claims

The supplementary results support three cautious claims. First, MAML-Select is not presented as a universal accuracy winner; it is competitive with FedAvg and FedGCS while reducing modelled resource use. Second, the

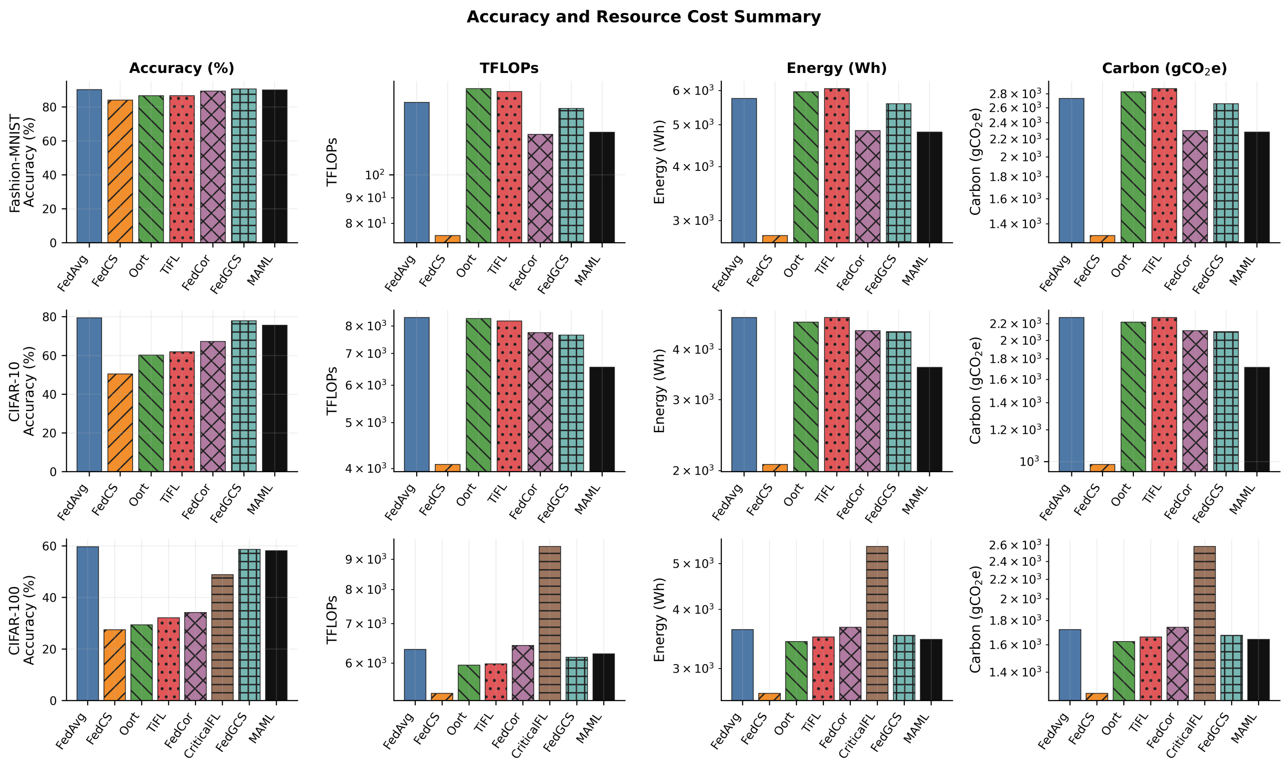


Figure 2: Expanded accuracy/resource grid across datasets and methods. This plot is retained as supplementary evidence because the main letter reports the same information more compactly through the benchmark table and trade-off figure.

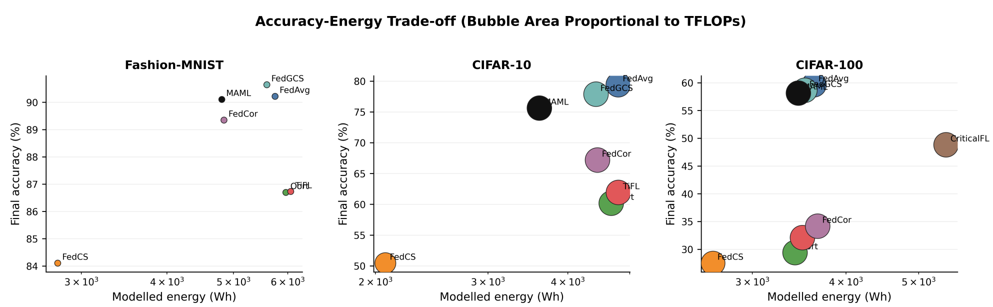
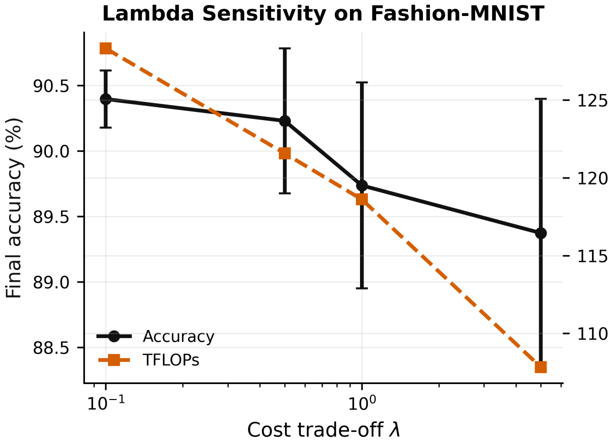


Figure 3: Accuracy–energy trade-off view used as a cross-check for the final normalized trade-off plot. The main paper uses the normalized compute view because communication is identical across selectors and the TFLOP axis is easier to compare across datasets.

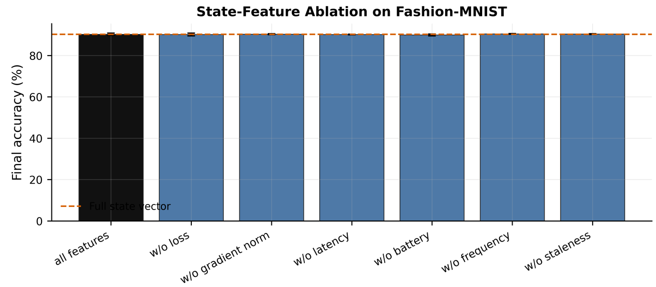
Table 2: Paired tests for MAML-Select against FedAvg over shared seeds ($n = 3$). Negative signed differences mean MAML-Select is lower than FedAvg.

Dataset	ΔAcc.	p	ΔTFLOPs	p	ΔEnergy	p	ΔCarbon	p
Fashion-MNIST	-0.11 pp	0.840	-17.93	0.009	-943.6 Wh	0.007	-448.2 g	0.007
CIFAR-10	-3.81 pp	0.041	-1792.16	0.002	-1187.4 Wh	0.001	-564.0 g	0.001
CIFAR-100	-1.51 pp	0.039	-107.80	0.012	-169.6 Wh	0.003	-80.6 g	0.003

resource estimates follow the stated simulator assumptions for device tiers and grid intensity. Third, fairness is reported explicitly: MAML-Select reaches full client coverage, but its Jain index is lower than FedAvg and FedGCS, so the manuscript avoids claiming fairness superiority.

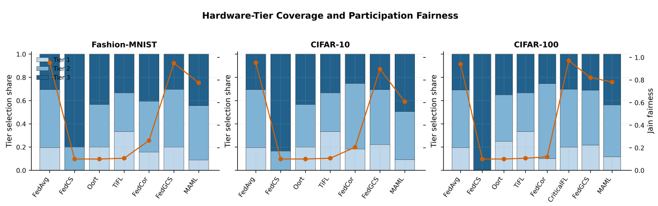


(a) λ sensitivity.

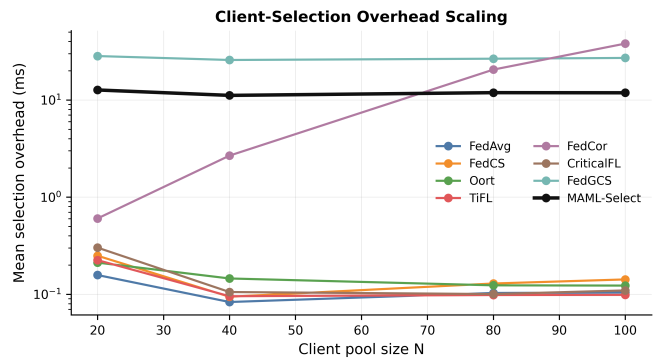


(b) State-feature ablation.

Figure 4: Design-analysis plots on Fashion-MNIST. The selected operating point is $\lambda = 0.5$. Accuracy remains stable over the sweep, while larger λ reduces compute, energy, and carbon at the cost of participation fairness.



(a) Fairness and tier coverage.



(b) Selector overhead scaling.

Figure 5: Reviewer-requested fairness and scalability checks. MAML-Select reaches full client coverage in the main runs, but its Jain index is below FedAvg and FedGCS; therefore the revised paper reports fairness transparently and does not claim fairness superiority. Selection overhead remains negligible relative to local training.

Table 3: λ sensitivity on Fashion-MNIST ($n = 3$). Accuracy is reported as mean $\pm$ s.d.; remaining columns are means.

λ	Acc. (%)	TFLOPs	Energy (Wh)	Carbon (g)	Jain
0.1	90.40 ± 0.22	128.3	5176	2459	0.91
0.5	90.23 ± 0.55	121.6	4799	2280	0.78
1.0	89.74 ± 0.79	118.6	4639	2204	0.71
5.0	89.37 ± 1.02	107.8	4140	1966	0.50

Table 4: State-feature ablation on Fashion-MNIST ($n = 3$). Each row removes one feature from the six-dimensional MAML-Select state vector.

Variant	Acc. (%)	TFLOPs	Energy (Wh)	Jain
Full state vector	90.23 ± 0.55	121.6	4799	0.78
w/o loss	90.16 ± 0.66	122.8	4851	0.79
w/o gradient norm	90.30 ± 0.24	122.4	4831	0.78
w/o latency	90.14 ± 0.14	118.9	4688	0.70
w/o battery	89.90 ± 0.44	122.2	4830	0.79
w/o frequency	90.47 ± 0.13	115.0	4533	0.65
w/o staleness	90.37 ± 0.19	122.7	4838	0.78

Table 5: Mean client-selection overhead in seconds for the scaling probe (10 rounds per setting).

Method	$N = 20$	$N = 40$	$N = 80$	$N = 100$
FedAvg	0.000158	0.000083	0.000103	0.000104
FedCor	0.000604	0.002676	0.020562	0.038099
FedGCS	0.028355	0.025831	0.026627	0.027116
MAML-Select	0.012689	0.011160	0.011907	0.011871